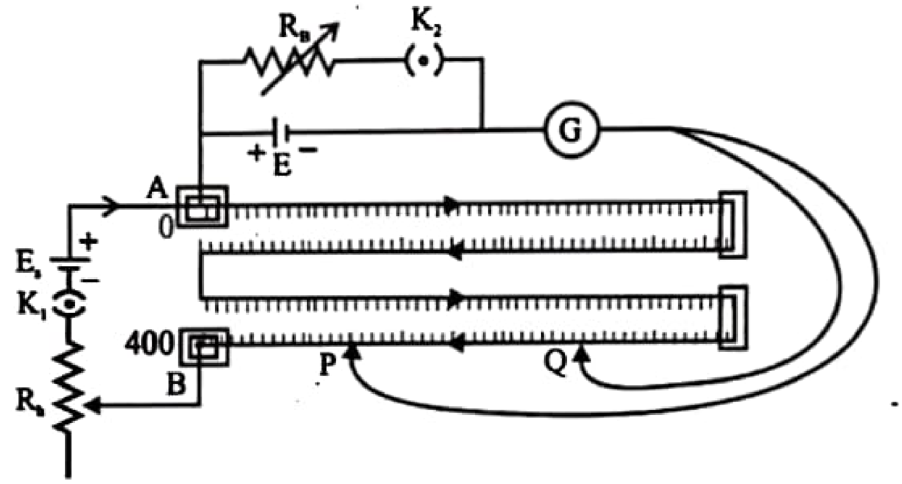
EXPERIMENT NO. 11
INTERNAL RESISTANCE OF A CELL

Aim: To determine the internal resistance of a given cell using potentiometer.

Apparatus: Potentiometer, rheostat, resistance box, battery, cell, jockey, plug key, galvanometer etc.

Diagram:



Formula:

$$r = \frac{(L_1 - L_2)S}{L_2}$$

Where, S is the shunt resistance in parallel with the given cell, L_1 and L_2 are balancing length without and with shunt respectively and r is the internal resistance of the cell.

Procedure:

1. Measure the e.m.f. (E_S) of the battery and the e.m.f. (E) of the cells. See that $E_S > E$.
2. Make the connections as shown in the figure above. Close the key K to bring the circuit in working. Adjust the rheostat at minimum resistance for maximum current in the circuit.
3. Then touch the jockey at zero end of the potentiometer wire and note the direction of deflection in the galvanometer. Touch the jockey at the other end of the potentiometer wire. If the direction of deflection is opposite to that in the first case, the connections are correct. (If the deflection is in the same direction then either connections are wrong or e.m.f. of the cell used is less).
4. First keep K_2 open and K_1 closed. Starting from the point A touch the jockey on the wire at various points and obtain the null point P_1. Note the length L_1 of the potentiometer wire between the point A and P_1.
5. Now, close K_1 as well as K_2 and keep suitable resistance S (10 Ω to 20 Ω) from resistance box. Again touch the jockey from point A on the wire at various points and obtain the null point P_2. Note the length L_2 of the potentiometer wire between the point A and P_2.
6. Take two more readings by changing the value of S.

Observation:

Balancing length when K_2 is open L_1 =...322... cm

Observation Table:

Obs. No.	Resistance in resistance box R_B (Ω) shunt	Balancing length L_2 (cm)	Internal resistance of a cell r (Ω)
1.	2	219	0·941
2.	3	222	1·351
3.	4	231	1·576
4.	5	243	1·625
			Mean 1·373

Calculations:

$$r = \frac{(L_1 - L_2)S}{L_2} = \frac{(322-219)2}{219} = 0{\cdot}941$$

$$= \frac{(322-222)3}{222} = 1{\cdot}351$$

$$= \frac{(322-231)4}{231} = 1{\cdot}576$$

$$= \frac{(322-243)5}{243} = 1{\cdot}625$$

Result:

Internal resistance of the given cell, r = 1·373 Ω

Precautions:

1. Once the rheostat is adjusted, the setting of the rheostat should not be changed throughout the experiment.
2. Current should be passed for short time.
3. See that $E_s > E$.

Questions

1. What do you mean by the internal resistance of cell?

The resistance offered by the electrolyte and other parts inside a cell to the flow of electric current is called the internal resistance of the cell.

2. What are the factors on which internal resistance of cell depends?

The distance between the electrodes, The area of the electrodes immersed in the electrolyte The nature and concentration of the electrolyte, The temperature of the electrolyte

3. Does internal resistance of cell depends on the current drawn from the cell?

No, the internal resistance of a cell does not depend on the current drawn from the cell. It depends on the physical characteristics of the cell and the electrolyte.

Remark and sign of teacher : ....